

Ahmedabad University

Neural Graphical Models for Healthcare Data: Representation, Learning, and Probabilistic Inference

g1_apl_team

Motivation Behind the Project

- **Why This Problem:**
  - Lung-cancer risk emerges from an interacting network of behavioral, demographic, and symptomatic factors — not any single variable in isolation
  - Standard classifiers predict yes/no well, but treat the process as a black box
  - Clinicians need to see **why** a prediction was made, not just the prediction itself.

- **Course Perspective:**
  - Apply Representation, Inference, and Learning to a real dataset, not just theory
  - Compare an undirected Conditional-Independence graph against a directed Bayesian Network on the same data
  - Use exact inference to answer partial-evidence clinical queries that a fixed classifier can only approximate via imputation
  - Ground every design choice against a recent research paper (SOTA positioning)

Real-World Perspective

Why This Problem Matters

- Clinical decision support benefits from models clinicians can inspect and reason about, not just point predictions
- Understanding which symptoms are conditionally dependent given smoking status (or age) highlights redundant vs. informative screening questions
- PGMs support partial-evidence queries — e.g. “what is P(lung cancer) given only 3 of 15 observed symptoms?” — which fixed-input classifiers cannot answer without retraining or imputation
- Interpretable risk modeling is directly transferable to other screening domains beyond lung cancer

Problem Statement

- **Objective:** Represent lung-cancer risk factors and symptoms as a Probabilistic Graphical Model, learn its structure and parameters from data, and support exact inference under partial evidence.

- **Scope Commitment (Applied AI Track):**
  - Domain: clinical / healthcare risk prediction
  - Principal dataset: Kaggle Lung Cancer Survey Dataset
  - 15 categorical/binary symptom & behavioral features + AGE
  - CDC infant-mortality data (used in the SOTA paper) is explicitly out of scope — too large for a one-semester project

- **Assumptions:**
  - Self-reported symptom labels are a reasonable, if noisy, proxy for clinical presentation
  - AGE is treated as continuous; all other features as binary/categorical
  - Label-noise analysis is deferred as a stated limitation, not ignored

PGM Mapping — Representation

- Random variables: 15 binary/categorical risk factors & symptoms + AGE + target LUNG_CANCER
- Two complementary graph structures recovered from the same data:

Structure	Method	What It Captures
Conditional-Independence (CI) graph	Sparse precision-matrix estimation (graphical lasso)	Undirected, symmetric dependence — which variables co-vary regardless of ordering
Bayesian Network (DAG)	Score-based structure learning (hill-climbing + BIC)	A candidate generative ordering, e.g. SMOKING → CHRONIC DISEASE → LUNG_CANCER

Comparing both structures on the same data is itself a Representation KPI — it tests how sensitive downstream inference is to the choice of graph.

PGM Mapping — Learning

- Structure learning: hill-climbing with BIC score (Bayesian Network) and graphical-lasso sparse precision estimation (CI graph)
- Parameter learning: Maximum Likelihood / Bayesian (Dirichlet-prior) estimation of conditional probability tables given the learned DAG
- Stretch extension: fit a small MLP-based Neural Graphical Model on the recovered graph, jointly optimizing regression fit and a soft structure-matching penalty — testing whether a richer functional form improves calibration without sacrificing the interpretability of a fixed graph

PGM Mapping — Inference

- **Target Queries**
  - P(LUNG_CANCER = Yes | observed symptom subset) — the primary clinical query under partial evidence
  - P(SMOKING | remaining features) — a secondary diagnostic query testing whether the model robustly reconstructs any variable, not only the labeled outcome
- **Method:**
  - Exact inference via variable elimination on the learned Bayesian Network for point and marginal queries
  - Stretch goal: gradient-based MAP inference (as used by our SOTA reference) on a small NGM-style network sharing the same graph, to compare inference behavior against the classical BN.

State-of-the-Art Position

Neural Graphical Models — Shrivastava & Chajewska, Microsoft Research (arXiv:2210.00453)

Field	Details
Problem	Represent arbitrary (directed / undirected / mixed-edge, mixed-datatype) feature-dependency graphs with one neural-network-based PGM
Representation	Each feature is a function of its graph neighbors (or Markov blanket), re-parameterized as an MLP constrained by the graph's zero pattern
Learning	Joint multi-task regression with a soft structure-matching penalty; proximal initialization warm-starts the weights
Inference	Message passing, and a gradient-based MAP update over unobserved input values
Evaluation	Synthetic GGMs, a ~300-patient lung-cancer dataset, and a 4M-record CDC infant-mortality dataset
Relation to our project	Evaluates on the same Lung Cancer dataset we use — our classical PGM baselines are directly comparable to its reported numbers

Baseline & Initial KPIs

Success is measured across all three PGM verticals — not accuracy alone

Vertical	KPI	Measured Against
Representation	Graph density & agreement between CI graph and BN; edge stability under bootstrap resampling	Cross-method agreement; domain sanity check
Inference	Log-likelihood of held-out patients; calibration (Brier, ECE); query latency	Naive Bayes; Logistic Regression
Learning	5-fold CV AUROC / AUPR / accuracy; sample efficiency (20%–100% of data)	Logistic Regression (matches SOTA paper's protocol), Naive Bayes

- **Baseline:** Logistic Regression (matching the SOTA paper's exact protocol for this dataset) and Gaussian Naive Bayes, a second PGM-flavored baseline assuming full feature independence given the class

Experimental Plan

Phase	Focus
Phase 1 (M1)	Data acquisition & cleaning, exploratory analysis, CI-graph and BN structure recovery, baseline implementation
Phase 2 (M2)	Parameter learning on recovered graphs, exact inference implementation, KPI evaluation, initial SOTA comparison
Phase 3 (M3)	Sensitivity analysis (CI graph vs. BN graph), calibration analysis, optional NGM-style neural-view extension
Phase 4 (Final)	Consolidation, ablations, writeup, and reflection against the SOTA position

Team Responsibilities

Member	Primary Responsibility
Priyam Modi	Problem formulation and overall PGM mapping; CI-graph and Bayesian Network structure & parameter learning; exact-inference design; SOTA positioning and analysis; KPI design across all verticals; code review and integration; report and video compilation
Rushin Modi	Data acquisition and cleaning; exploratory data analysis; dataset documentation; Logistic Regression baseline
Jash Odedra	Naive Bayes baseline; KPI computation and metrics logging; video recording support

Timeline

Week	Milestone	Deliverable
1–2	Data cleaning + CI graph + BN structure learning	Draft graphs
3–4	Parameter learning + exact inference + baselines	M2 report
5–6	KPI evaluation + sensitivity analysis	Evaluation notebook
7–8	Optional NGM-style extension + calibration study	M3 report
9	Consolidation, ablations, final writeup	Final submission

Where We Stand at M1

M1 Objective	Status
Problem statement, domain & dataset commitment	Defined — clinical risk prediction, Kaggle Lung Cancer dataset
PGM mapping (Representation / Inference / Learning)	Fully scoped for both CI graph and Bayesian Network
SOTA position	Identified & analyzed — Neural Graphical Models (Shrivastava & Chajewska, 2022)
Baseline & KPIs	Defined across all three PGM verticals, not accuracy alone
Experimental plan, responsibilities, timeline	Laid out through Final submission

References

H. Shrivastava and U. Chajewska, “Neural Graphical Models,” Oct. 2022. arxiv.org/abs/2210.00453v4

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